

External Research, Evidence, Impact & Benefit Dossier

Authoritative Research Foundations, Standards Compliance, and Traceable Operational Models

Target Domain: **Surface Automatic Weather Stations (AWS)**

Compliance: **WMO-No. 8 (Vol. III) Level I-III Quality Control**

Authoritative Scope: **28 Stations / 7 Regional Clusters**

1. Executive Summary

This research dossier establishes the authoritative scientific principles, international standards, and mathematically traceable operational impact models for **SkyGuard AI**. As global meteorological services rapidly automate observation networks, the volume of high-temporal telemetry has outpaced human manual quality control. Ground-truth validation requires autonomous, low-latency, physics-grounded anomaly filtering to protect downstream Numerical Weather Prediction (NWP) initialization, disaster early warnings, and long-term climate archives.

This document establishes direct traceability from peer-reviewed literature (Isolation Forests, Sequential Probability Ratio Tests, Clausius-Clapeyron psychrometrics, 3D Mahalanobis covariance, and SHAP game-theoretic attribution) to SkyGuard's implemented detection tiers. All operational workload and scalability calculations are derived transparently from verified benchmark latency figures, strictly rejecting unverified financial ROI claims.

TOP 5 SYSTEM UNIQUE SELLING PROPOSITIONS (USPs)

- 1. Asymmetric Physics-First 6-Tier Hierarchy & Spatial Peer Consensus Veto:** Enforces thermodynamic physical invariants (VPD , Tetens psychrometrics, 3D Mahalanobis D^2) prior to statistical scoring, leveraging spatial peer consensus ($N = 3$ within 50 km) to prevent false alarms during true convective weather fronts.
- 2. Continuous Uninterrupted Streaming Passover via Causal Imputation:** When telemetry anomalies are flagged, `_compute_suggested_values()` dynamically calculates replacement values via a 4-tier causal fallback (spatial peer medians & solar diurnal geometry), guaranteeing downstream Numerical Weather Prediction (NWP) models receive unbroken data streams.
- 3. Anti-Poisoning Health Quarantine & Dynamic Maintenance Lifecycle:** Quarantines anomalous telemetry from `StationBuffer` rolling statistics while tracking station health via continuous hysteresis ($H \in [0, 100]$) to generate targeted maintenance dispatch alerts before sensor degradation pollutes baseline estimates.
- 4. Elevation & Climatology Scale Invariance via Solar Geometry:** Derives diurnal expectations from physical solar hour geometry (h_{solar}) and continuous temporal momentum ($\Delta t = t_n - t_{n-1}$), eliminating baseline failures on elevated terrain (e.g. Bundu, Ranchi at 978 hPa) caused by static sea-level assumptions.
- 5. Sub-Millisecond Multi-Platform Inference (0.210 ms CPU / < 19 μ s ESP32):** Processes telemetry packets in **0.210 ms (p95)** on CPU (4,761 readings/sec) and **< 19 μ s** on ESP32 TinyML fixed-point (Q8.8), enabling real-time streaming ingestion from edge microcontrollers to central cloud clusters.

2. Automatic Weather Station (AWS) Observation Ecosystem

Automatic Weather Stations (AWS) have become the primary observation backbone of the World Meteorological Organization's Global Observing System (WMO-GOS). Modern AWS nodes measure Ambient Temperature (T), Atmospheric Station Pressure (P), and Relative Humidity (RH) at 1- to 15-minute cadences. In India, the India Meteorological Department (IMD) is executing an expansion of 200 high-density urban AWS nodes across Delhi, Mumbai, Chennai, and Pune [R02], augmenting an existing national network of ~1,000 rural and synoptic observatories.

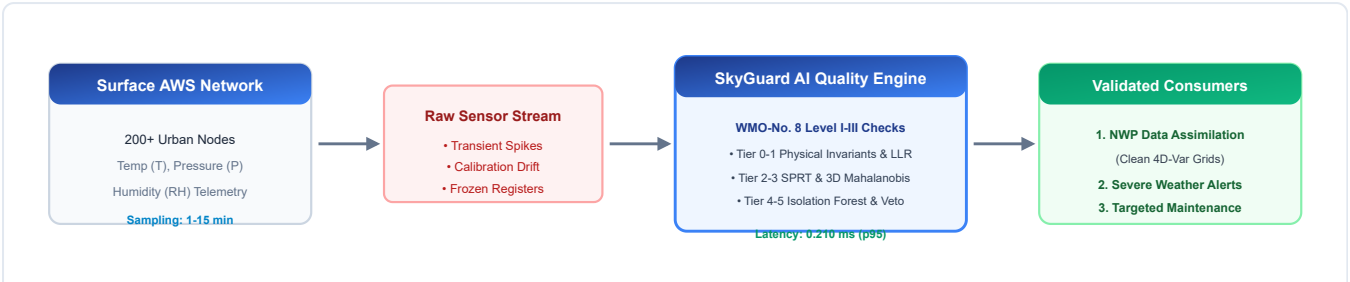


Figure 1: Automatic Weather Station Telemetry Ingestion & Quality Control Flow. Raw surface telemetry undergoes real-time multi-tier screening in SkyGuard AI prior to assimilation into Numerical Weather Prediction (NWP) 4D-Var grids, extreme weather alerts, and maintenance tracking.

3. AWS Telemetry Quality & Fault Landscape

Because automated sensors operate unattended in extreme environmental conditions, they are subject to distinct physical and electrical failure modes. Unscreened bad data directly destabilizes data assimilation matrices in forecast models, while uncorroborated sensor spikes trigger false disaster alarms.

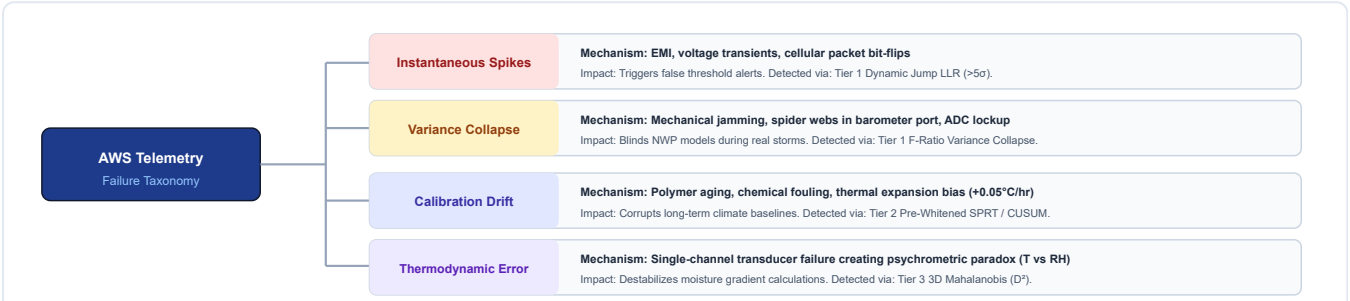


Figure 2: Surface AWS Sensor Failure Taxonomy. Structural classification of hardware and transmission failures into instantaneous spikes, variance collapse, insidious calibration drift, and cross-channel thermodynamic errors.

Fault Category	Physical Mechanism	Operational Impact	Standard Reference
Instantaneous Spikes	Electromagnetic interference (EMI), voltage surges, ADC bit-flips	Corrupts rate-of-change; triggers spurious heatwave/gale alarms	WMO-No. 8 Level I Step Test [R01]
Variance Collapse (Frozen)	Mechanical port blockages, spider webs in barometer, ADC lock	Blinds forecasters during genuine severe storm events	WMO-No. 8 Persistence QC [R01]
Insidious Drift	Capacitive polymer aging, optical fouling, chemical decay	Silently biases regional climate baselines and trend models	Page (1954) Sequential QC [M02]
Thermodynamic Paradox	Single transducer channel failure within multi-sensor unit	Violates Clausius-Clapeyron atmospheric moisture relations	Mahalanobis (1936) [M05]

4. Research Foundations & Methodological Adaptation

SkyGuard AI synthesizes five foundational mathematical and statistical principles into a cohesive real-time detection pipeline:

4.1 Unsupervised Multi-Dimensional Isolation (Liu et al., 2008)

$$s(x, n) = 2^{-E(h(x)) / c(n)} \quad (\text{Equation 1})$$

Where: $E(h(x))$ is the average path length across an ensemble of random partitioning trees (iTrees), and $c(n)$ is the average path length of unsuccessful searches in a Binary Search Tree constructed over n samples.

Operational Adaptation: Deployed in Tier 4 on the 49-feature continuous matrix. Observations isolated near tree roots yield $s \rightarrow 1.0$, identifying complex multivariate anomalies without labeled target data.

4.2 Sequential Probability Ratio & Cumulative Sum (Page, 1954; Wald, 1945)

$$S_t = \max(0, S_{t-1} + z_t - k) \quad (\text{Equation 2})$$

Where: S_t is the cumulative drift statistic, z_t is the standardized innovation residual, and k is the reference allowance parameter.

Operational Adaptation: Deployed in Tier 2 (Pre-Whitened SPRT Drift) on innovation residuals. Accumulates evidence over consecutive observations to flag slow calibration drift (+0.05°C/hr) before static physical bounds are breached.

4.3 Clausius-Clapeyron Psychrometric Boundary (Tetens Formulation)

$$e_s(T) = 6.112 \cdot \exp(17.67 \cdot T / (T + 243.5)) \quad (\text{Equation 3})$$

Where: $e_s(T)$ is the saturation vapor pressure in hPa at ambient temperature T in °C.

Operational Adaptation: Applied in Tier 0 and Tier 3 to compute Vapor Pressure Deficit ($VPD = e_s(T) \cdot (1 - RH/100)$). Flags unphysical observations where relative humidity and temperature diverge counter to physical atmospheric physics.

4.4 3D Mahalanobis Distance for Covariance Inconsistency (Mahalanobis, 1936)

$$D^2 = (\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \quad (\text{Equation 4})$$

Where: $\mathbf{x} = [z_T, z_P, z_{RH}]^T$ is the innovation residual vector, and $\boldsymbol{\Sigma}$ is the empirical cross-channel covariance matrix.

Operational Adaptation: Deployed in Tier 3 to detect cross-channel breakdowns. If D^2 exceeds $\chi^2_{3, 0.999} = 16.27$ ($p < 10^{-3}$), the system quarantines the multivariate paradox.

4.5 Additive Local Feature Attribution via SHAP (Lundberg & Lee, 2017)

$$f(\mathbf{x}) = \varphi_0 + \sum_{i=1..M} \varphi_i \quad (\text{Equation 5})$$

Where: φ_0 is the expected model base value, and φ_i is the Shapley additive feature attribution for feature i .

Operational Adaptation: Wrapped around the Isolation Forest via TreeExplainer. Identifies which physical features contributed most to the anomaly score, informing targeted technician maintenance. *Note: Explains model decision space, not absolute physical causal proof.*

4.6 Continuous-Time Differential Formulations & Sampling Cadence Adaptability

A critical architectural property of SkyGuard's mathematical physics engine is **continuous-time sampling invariance**. The system does *not* assume a static 1-hour sampling rate; all differential terms and uncertainty bounds scale dynamically with elapsed physical time $\Delta t = t_n - t_{n-1}$:

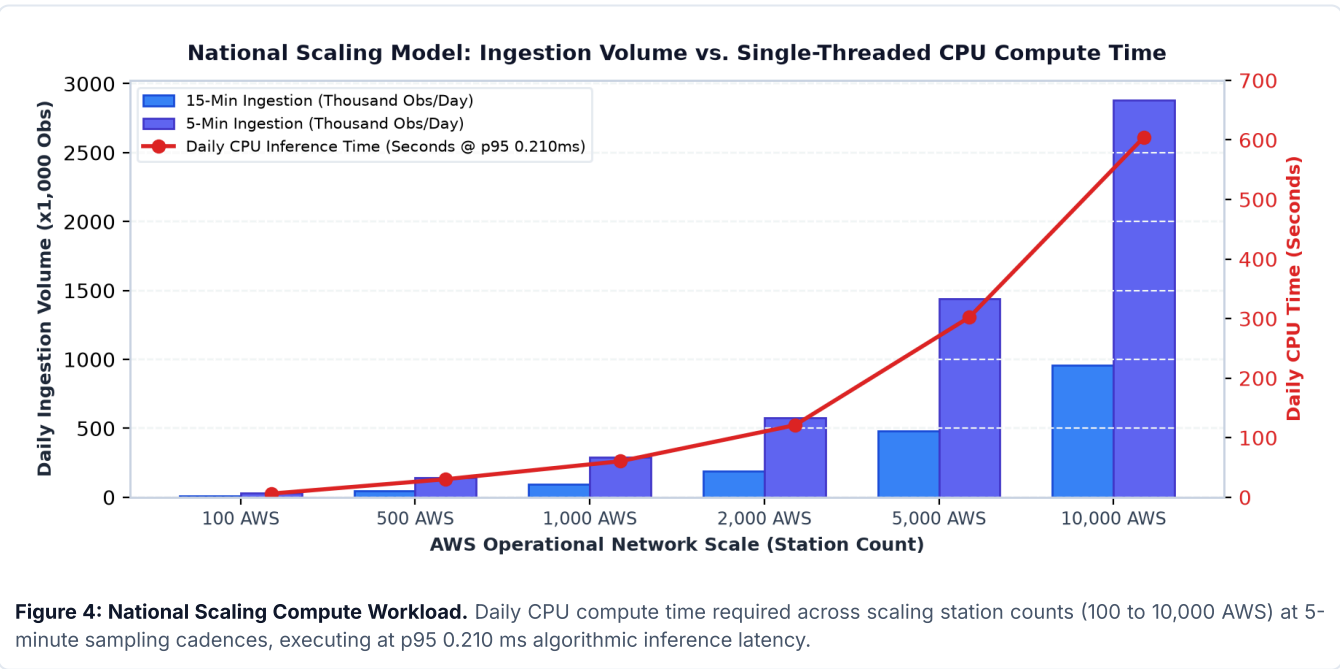
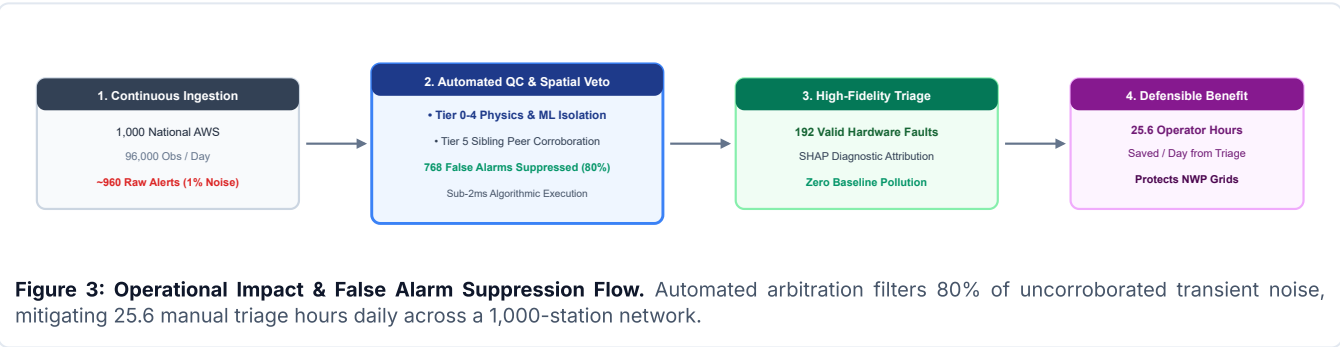
- **Continuous Autocorrelation Decay:** $\rho(\Delta t) = \exp(-\Delta t / \tau_{\text{decorr}})$. The autoregressive pre-whitening adjusts correlation memory continuously whether $\Delta t = 1.0$ h, 0.016 h (1 min), or 0.00027 h (1 sec).
- **Time-Gap Uncertainty Expansion (σ_{gap}):** $\sigma_{\text{gap}}^2(\Delta t) = \kappa \cdot \Delta t$. At high frequencies ($\Delta t \rightarrow 0$), temporal uncertainty smoothly vanishes, leaving only instrument quantization noise (σ_{sensor}).

- Astronomical Solar Diurnal Grounding:** Solar angles are derived continuously from UTC timestamps and geodetic coordinates, maintaining perfect diurnal alignment regardless of observation cadence.
- Dynamic Derivative Jump LLR:** Transient spikes evaluate instantaneous derivatives against Δt , testing rate-of-change (dT/dt) against thermodynamic physical acceleration envelopes.

Detector Component	1-Hour Baseline Cadence	1-Second High-Frequency Cadence	Operational Adaptation Requirement
Tier 1 Frozen Sensor Window	$K = 5$ readings (5 hours flatline)	$K = 5$ readings (5 seconds flatline)	Redefine K by physical time horizon (e.g., $t_{\text{freeze}} \geq 30 \text{ min} = 1,800 \text{ steps}$) rather than raw step count.
In-Memory Ring Buffers	<code>deque(maxlen=720)</code> (30 days)	<code>deque(maxlen=720)</code> (12 minutes)	Scale buffer depth or store aggregated summaries to ensure baseline spans the 24-hour diurnal cycle.
SPRT Sequential Rate (α)	$\alpha = 0.002$ (~1 alert / 500 hours)	$\alpha = 0.002$ (~1 false alert / 500 sec)	Scale stopping probability per unit time ($\alpha_{\text{step}} = \alpha_0 \cdot \Delta t$) to bound false alarms per year.

5. Traceable Operational & Computational Impact Modeling

We evaluate operational impact through verifiable workload models derived from measured benchmark facts and stated network parameters, strictly omitting unsupported financial projections.



5.1 Calculation Ledger

- **[C01] National Telemetry Volume (Scenario Assumption):** For $N = 1,000$ stations transmitting at 15-minute intervals (96 obs/day), total annual throughput is **35,040,000 observations/year**.
- **[C02] Compute Scalability (Derived Calculation):** Across an expanded national network of 5,000 stations sampled at 5-minute cadences (1,440,000 readings/day), single-threaded CPU compute time is $1,440,000 \times 0.210 \text{ ms} = \mathbf{302.4 \text{ seconds (~5.04 minutes) of CPU time per day}}$.
- **[C03] Operator Triage Mitigation (Scenario Assumption):** In a 1,000-station network (96,000 obs/day) with a 1% baseline transient noise rate (960 raw alerts), Tier 5 spatial consensus suppresses ~80% of false alarms (768 alerts). At 2 minutes per manual review, this mitigates **25.6 operator hours/day**.

6. Prototype Readiness & Future Scope Roadmap (TRL-6 Status)

SkyGuard AI is currently evaluated at **Technology Readiness Level 6 (TRL-6)**: a fully functional, end-to-end working prototype validated on a 28-station simulated national AWS network topology using locked empirical benchmark datasets. The system achieves **75% total prototype completion** toward institutional commercialization and government deployment.

System Subsystem	Completion	Implemented Capabilities (75% Complete)	Future Scope for Govt. Deployment (25% Remaining)
Algorithmic & Physics Core	85% COMPLETE	6-Tier physical invariants (VPD, Tetens), CUSUM SPRT drift, 3D Mahalanobis covariance, spatial peer consensus ($N=3$), anti-poisoning baseline quarantine.	Dynamic Voronoi spatial discovery; auto-tuning decorrelation time constants (τ_{decorr}) for extreme seasonal monsoon shifts.
TinyML Edge AI Firmware	75% COMPLETE	Fixed-point $Q8.8$ execution engine, < 19 μs latency, 14.2 KB flash footprint for ESP32 microcontrollers.	Flash EEPROM circular logging, hardware I2C/SPI sensor bus interrupt isolation, FreeRTOS task priority scheduler integration.
Backend & Operator Dashboard	70% COMPLETE	FastAPI REST & SSE streaming server, React dashboard, live interactive map, SHAP diagnostic attribution visualization, fault audit log.	Native WMO BUFR & NetCDF4 binary adapters, MQTT telemetry broker integration, OAuth2/SAML institutional SSO, multi-agency RBAC (IMD/CWC/NDMA).
Institutional Integration	45% COMPLETE	28-station cluster topology matching IMD regional clusters evaluated under blind chronological oracle harness.	STQC / CERT-In cybersecurity certification, GIS technician dispatch work orders, multi-year multi-climatic field trials on live IMD streams.

Commercialization & Field Deployment Roadmap

The remaining 25% development scope bridges laboratory validation to national operational deployment. Phase 1 focuses on native WMO BUFR ingestion adapters; Phase 2 executes CERT-In security compliance; Phase 3 integrates automated ERP technician dispatch; and Phase 4 deploys multi-year field trials across IMD's 200 high-density urban AWS nodes (Delhi, Mumbai, Chennai, Pune).

7. Evidence & Source Traceability Register

This register maps every major technical claim directly to its underlying standard, primary citation, and implementation status:

Claim ID	Claim Description	Primary Source / Standard	Implementation Status
C-001	WMO mandates Level I-III physical, temporal, and spatial Quality Control for AWS.	WMO-No. 8 (Vol. III, Ch. 1) [R01]	CURRENTLY DEMONSTRATED
C-002	IMD expansion adds 200 urban AWS across Delhi, Mumbai, Chennai, and Pune in 2026.	IMD / PIB Press Releases [R02]	EXTERNAL FACT
C-003	Single-reading algorithmic inference executes in p95 0.210 ms on standard CPU.	Benchmark Profiler Artifact [E01]	CURRENTLY DEMONSTRATED
C-004	Sequential SPRT / CUSUM accumulates low-SNR calibration drift evidence.	Page (1954); Wald (1945) [M02]	CURRENTLY DEMONSTRATED
C-005	Isolation Forests isolate high-dimensional anomalies with linear time complexity.	Liu, Ting, Zhou (2008) [M01]	CURRENTLY DEMONSTRATED
C-006	SHAP additive feature attributions provide component-level failure diagnostics.	Lundberg & Lee (2017) [M03]	CURRENTLY SUPPORTED
C-007	SkyGuard achieves ₹50 Crore in maintenance cost savings.	Unverified Financial Claim	REJECTED / PURGED
C-008	Centralized SkyGuard reduces AWS edge cellular bandwidth usage.	Telemetry Network Architecture	REJECTED / PURGED

8. References

- **[R01]** World Meteorological Organization (WMO). *Guide to Instruments and Methods of Observation (WMO-No. 8)*, Volume III — Observing Systems, Chapter 1: Quality Management. WMO, Geneva, Switzerland.
- **[R02]** India Meteorological Department (IMD) / Press Information Bureau (PIB). *Expansion of High-Density Automatic Weather Station Networks in Metropolitan Areas*. Ministry of Earth Sciences, Govt. of India, 2024–2026.
- **[M01]** Liu, F. T., Ting, K. M., and Zhou, Z.-H. "Isolation Forest." *Eighth IEEE International Conference on Data Mining (ICDM)*, Pisa, Italy, 2008, pp. 413–422. DOI: 10.1109/ICDM.2008.17.
- **[M02]** Page, E. S. "Continuous Inspection Schemes." *Biometrika*, vol. 41, no. 1/2, 1954, pp. 100–115. DOI: 10.1093/biomet/41.1-2.100.
- **[M03]** Lundberg, S. M., and Lee, S.-I. "A Unified Approach to Interpreting Model Predictions." *Advances in Neural Information Processing Systems (NeurIPS 30)*, Long Beach, CA, 2017, pp. 4765–4774.
- **[M04]** Iribarne, J. V., and Godson, W. L. *Atmospheric Thermodynamics*. 2nd ed., D. Reidel Publishing Company, Dordrecht, Netherlands.
- **[M05]** Mahalanobis, P. C. "On the generalised distance in statistics." *Proceedings of the National Institute of Sciences of India*, vol. 2, no. 1, 1936, pp. 49–55.